Labsheet-10

Name : Kashish chauhan sec - B , Roll no : 39 , Mastery in Data Analytics and Visualizations and Career Advancement

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import pearsonr
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
from google.colab import files
uploaded = files.upload()
df = pd.read_csv(next(iter(uploaded)))
print(df.shape)
df.head()
```

Choose Files country_wis…olumns.csv

country_wise_latest-selected-columns.csv(text/csv) - 8918 bytes, last modified: 9/16/2026 - 100% done

```
Saving country_wise_latest-selected-columns.csv to
country_wise_latest-select (187, 10)
```

	Country/Region	Confirmed	Deaths	New deaths	recove
	Recovered	Active	New cases		
0	Afghanistan	36263	1269	25198	9796
106	10				
1	Albania	4880	144	2745	1991
117	6				
2	Algeria	27973	1163	18837	7973
616	8				
3	Andorra	907	52	803	52
10	0				
4	Angola	950	41	242	667
18	1				

Generate code with df New interactive sheet

Next steps:

```
df.columns = df.columns.str.strip()
print(df.columns.tolist())
df.info()
```

```
['Country/Region', 'Confirmed', 'Deaths', 'Recovered', 'Active', 'New
cases', <class 'pandas.core.frame.DataFrame'>
RangeIndex: 187 entries, 0 to 186
Data columns (total 10 columns):
# Column Non-Null Count Dtype
--- ------ -------------- -----
 0 Country/Region 187 non-null object
```

```
 1 Confirmed 187 non-null int64
 2 Deaths 187 non-null int64
 3 Recovered 187 non-null int64
 4 Active 187 non-null int64
 5 New cases 187 non-null int64
```



```
 6 New deaths 187 non-null int64
 7 New recovered 187 non-null int64
 8 Deaths / 100 Cases 187 non-null float64
 9 Recovered / 100 Cases 187 non-null float64
dtypes: float64(2), int64(7), object(1)
memory usage: 14.7+ KB
```

```
print(df.isnull().sum())
print("Duplicate Rows:", df.duplicated().sum())
df = df.drop_duplicates()
```

```
Country/Region 0
Confirmed 0
Deaths 0
Recovered 0
Active 0
New cases 0
New deaths 0
New recovered 0
Deaths / 100 Cases 0
Recovered / 100 Cases 0
dtype: int64
Duplicate Rows: 0
```

```
date_cols = [c for c in df.columns if "date" in c.lower()]
if date_cols:
df[date_cols[0]] = pd.to_datetime(df[date_cols[0]], errors="coerce")
print(df.describe())
```

```
 Confirmed Deaths Recovered Active New cases  count 1.870000e+02
187.000000 1.870000e+02 1.870000e+02 187.000000  mean 8.813094e+04
3497.518717 5.063148e+04 3.400194e+04 1222.957219  std 3.833187e+05
14100.002482 1.901882e+05 2.133262e+05 5710.374790  min 1.000000e+01
0.000000 0.000000e+00 0.000000e+00 0.000000  25% 1.114000e+03 18.500000
6.265000e+02 1.415000e+02 4.000000  50% 5.059000e+03 108.000000
2.815000e+03 1.600000e+03 49.000000  75% 4.046050e+04 734.000000
2.260600e+04 9.149000e+03 419.500000  max 4.290259e+06 148011.000000
1.846641e+06 2.816444e+06 56336.000000

 New deaths New recovered Deaths / 100 Cases Recovered / 100 Cases
count 187.000000 187.000000 187.000000 187.000000  mean 28.957219
933.812834 3.019519 64.820535  std 120.037173 4197.719635 3.454302
26.287694  min 0.000000 0.000000 0.000000 0.000000  25% 0.000000
0.000000 0.945000 48.770000  50% 1.000000 22.000000 2.150000 71.320000
75% 6.000000 221.000000 3.875000 86.885000  max 1076.000000
33728.000000 28.560000 100.000000
```

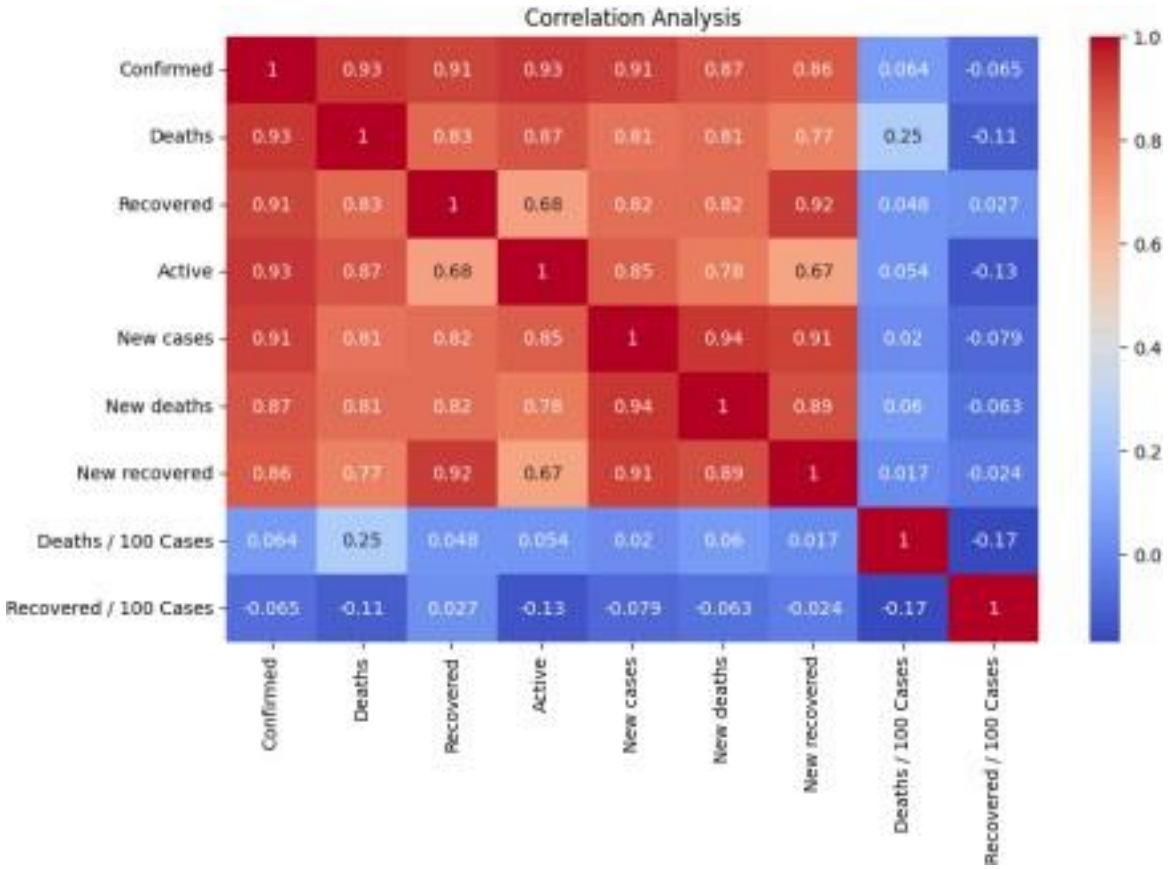
```
num_cols = df.select_dtypes(include=np.number).columns.tolist()
print("Numerical Columns:", num_cols)
```

```
Numerical Columns: ['Confirmed', 'Deaths', 'Recovered', 'Active', 'New
```

```
cases'
```



```
if len(num_cols) >= 2:
    plt.figure(figsize=(10,6))
sns.heatmap(df[num_cols].corr(), annot=True, cmap="coolwarm")
    plt.title("Correlation Analysis")
    plt.show()
```




```
if date_cols and num_cols:
    date_col = date_cols[0]
    value_col = num_cols[0]
    trend = df.groupby(date_col)[value_col].sum()
```

```
    plt.figure(figsize=(12,5))
    plt.plot(trend)
    plt.title("COVID-19 Trend Over Time")
    plt.xlabel("Date")
    plt.ylabel(value_col)
    plt.xticks(rotation=45)
    plt.show()


if len(num_cols) >= 2:
    x = df[num_cols[0]].dropna()
    y = df.loc[x.index, num_cols[1]].dropna()
    x, y = x.align(y, join="inner")

    correlation, p_value = pearsonr(x, y)
    print("Correlation:", correlation)
    print("P-value:", p_value)
```



```
Correlation: 0.9346984343393535
P-value: 4.8856399457711744e-85


data = df[num_cols].dropna()
X = data[[num_cols[0]]]
y = data[num_cols[1]]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)
```

▾ LinearRegression i ?

```
LinearRegression()


y_pred = model.predict(X_test)

mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)

print("MAE:", mae)
print("RMSE:", rmse)
print("R2 Score:", r2)

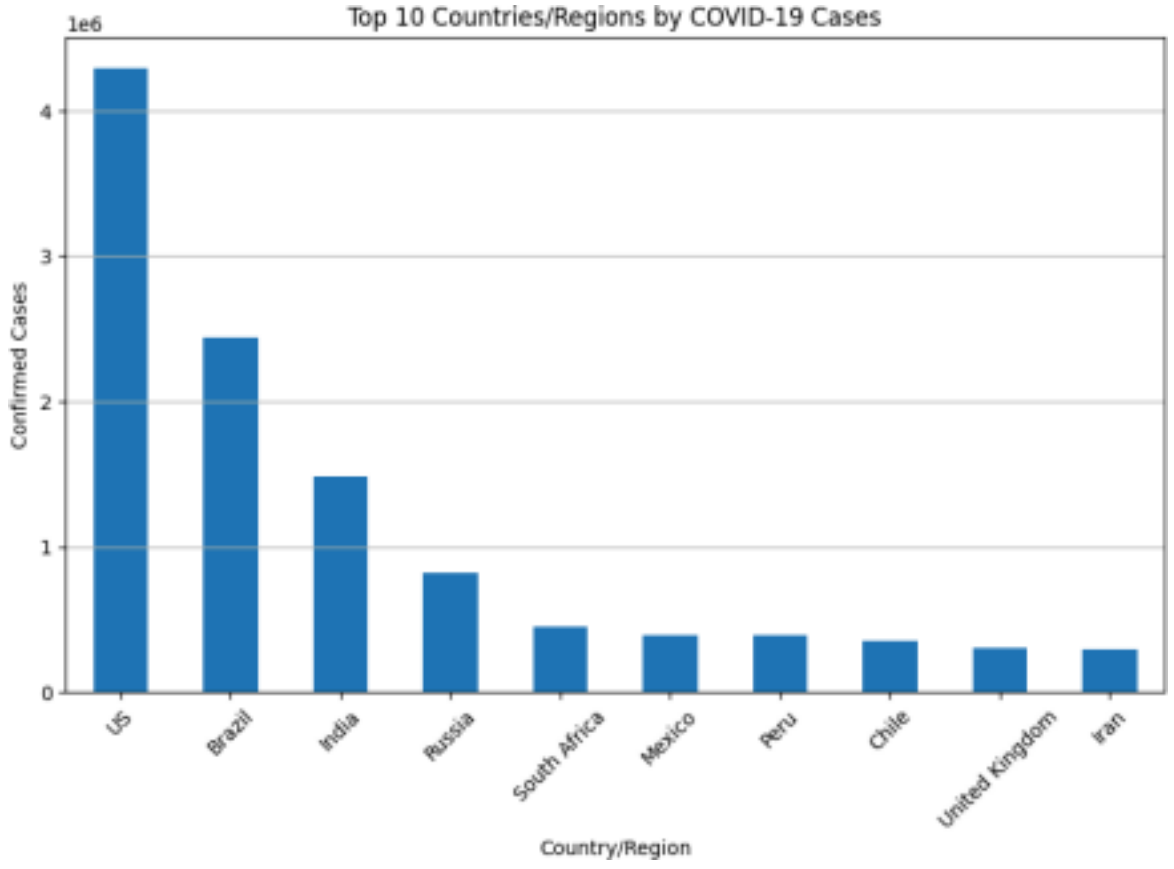
MAE: 3277.4754375596644
RMSE: 7901.046835773879
```

```
R2 Score: 0.27192084936935956


loc_col = next((c for c in df.columns if any(x in c.lower() for x in
["country case_col = next((c for c in df.columns if "confirm" in
c.lower() or "case" in

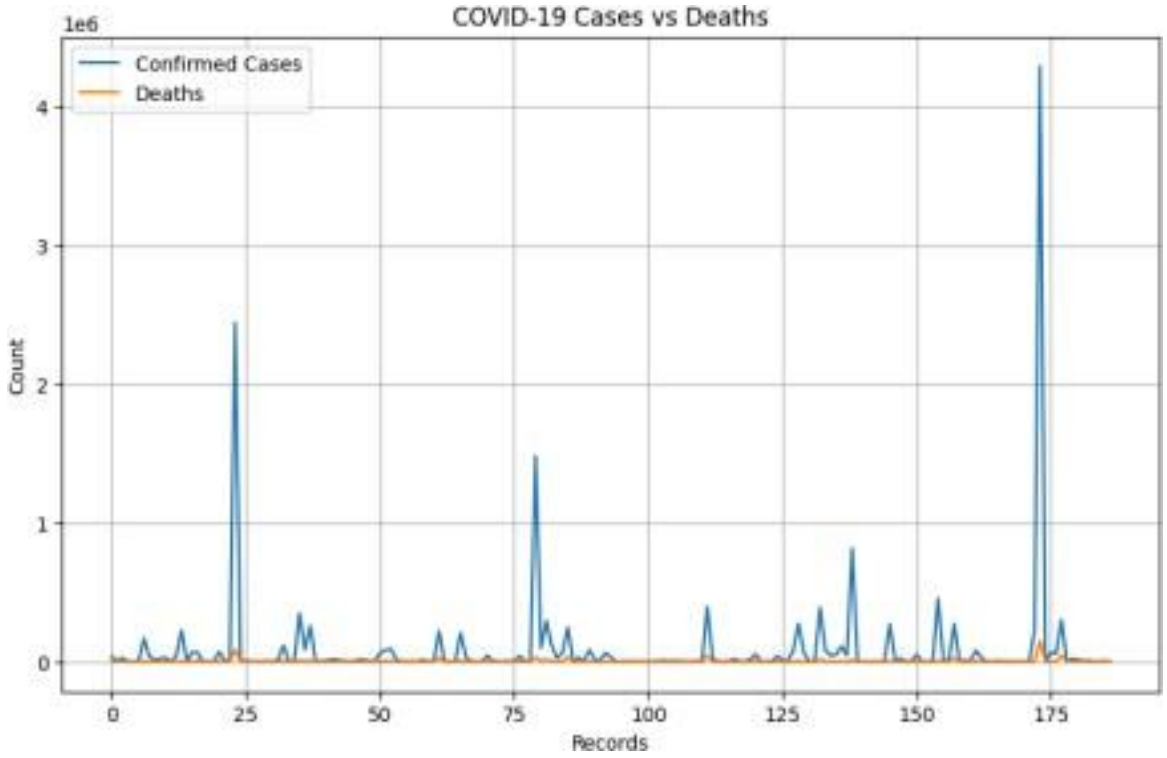
if loc_col and case_col:
top = df.groupby(loc_col)[case_col].sum().sort_values(ascending=False).hea

    plt.figure(figsize=(10,6))
    top.plot(kind="bar")
    plt.title("Top 10 Countries/Regions by COVID-19 Cases")
    plt.xlabel(loc_col)
    plt.ylabel("Confirmed Cases")
    plt.xticks(rotation=45)
    plt.grid(axis="y")
    plt.show()
```


```
case_col = next((c for c in df.columns if "confirm" in c.lower() or
"case" i death_col = next((c for c in df.columns if "death" in
c.lower()), None)
```

```
if case_col and death_col:
    plt.figure(figsize=(10,6))
    plt.plot(df[case_col].values, label="Confirmed Cases")
    plt.plot(df[death_col].values, label="Deaths")
    plt.title("COVID-19 Cases vs Deaths")
    plt.xlabel("Records")
    plt.ylabel("Count")
    plt.legend()
    plt.grid(True)
    plt.show()
```


```
plt.figure(figsize=(8,5))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.title("Actual vs Predicted COVID-19 Values")
plt.show()
```

```
        print("KEY INSIGHTS")
        print("1. COVID-19 trends can be analyzed over time.")
        print("2. Correlation analysis identifies relationships between
        numerical va print("3. Linear Regression can be used for basic
        prediction.")
        print("4. MAE, RMSE and R2 Score evaluate model performance.")
print("5. Analytics can support healthcare resource planning and decision-ma
```

Theory Questions & Answers

1. What are the major stages involved in an end-to-end data analytics project?

An end-to-end data analytics project consists of several stages. First, relevant data is collected from reliable sources. The collected data is then cleaned and preprocessed by handling missing values, duplicates, outliers, and categorical variables. After preprocessing, Exploratory Data Analysis (EDA) is performed to understand patterns and relationships in the data. Statistical analysis is used to identify significant relationships between variables. Visualization and dashboards are created to communicate the findings. After this, an appropriate machine learning model is developed and evaluated using suitable performance metrics. Finally, the results are converted into actionable business insights and presented through a report or dashboard.

2. Why is data preprocessing considered the foundation of successful analytics and machine learning?

Data preprocessing is important because real-world datasets are often incomplete, inconsistent, noisy, or incorrectly formatted. Preprocessing involves handling missing values, removing duplicate records, treating outliers, encoding categorical variables, and scaling numerical features when required. Poor-quality data can lead to inaccurate analysis and unreliable machine learning predictions. Proper preprocessing improves data quality and helps the model learn meaningful patterns. Therefore, preprocessing is considered one of the most important foundations of a successful analytics project.

3. How does Exploratory Data Analysis (EDA) contribute to model development?

EDA helps analysts understand the structure and characteristics of a dataset before developing a machine learning model. It uses descriptive statistics, graphs, correlation analysis, and distribution analysis to identify trends, patterns, outliers, and relationships between variables. EDA can also reveal irrelevant or highly correlated features. Based on these findings, appropriate features and algorithms can be selected. Thus, EDA helps improve the quality of feature selection, model design, and final predictions.

4. What factors should be considered while selecting a machine learning algorithm for a business problem?

The choice of a machine learning algorithm depends on several factors. First, the type of problem must be identified, such as classification, regression, or clustering. The size and structure of the dataset, number of features, type of variables, and presence of missing or noisy data should also be considered. Model accuracy, training time, interpretability, and computational requirements are important factors. Business requirements should also be considered because some businesses need highly interpretable models even if a more complex model provides slightly different results.

5. Why is model evaluation essential before deploying a predictive model?

Model evaluation is necessary to determine whether a predictive model performs well on unseen data. A model may perform very well on training data but poorly on new data due to overfitting. Evaluation helps identify such problems and measures the reliability of predictions. Different metrics are used for different problems. For classification, Accuracy, Precision, Recall, and F1-Score are commonly used, while regression models can be evaluated using MAE, RMSE, and R^2 Score. Proper evaluation helps ensure that the model is suitable for practical use.

6. Explain how Business Intelligence tools complement machine learning in data analytics projects.

Business Intelligence (BI) tools such as Tableau and Microsoft Power BI help present analytical results in an easy-to-understand format. Machine learning models can identify patterns and generate predictions, while BI tools can display these results using interactive charts, dashboards, and KPIs. For example, a predictive model may forecast COVID-19 cases, while a dashboard can display historical trends, predictions, and important indicators. This combination allows managers and decision-makers to understand analytical results and monitor important business or operational metrics.

7. What challenges are commonly encountered while working with real-world datasets?

Real-world datasets often contain many problems. Missing values may occur because information was not recorded properly. Duplicate records can affect the accuracy of analysis. Outliers can influence statistical calculations and machine learning models. Categorical data may need to be converted into numerical form. Different sources may use different formats and units. Large datasets can also require significant computational resources. In addition, data privacy, security, data quality, and class imbalance can create further challenges. Proper preprocessing and validation are required to handle these issues.

8. How can analytical insights be converted into actionable business recommendations?

Analytical insights become useful when they are connected to a specific business objective. First, important patterns or problems are identified from the data. Next, their possible impact on the organization is analyzed. Based on the findings, specific and measurable actions can be

recommended. For example, if COVID-19 analysis shows increasing cases in particular regions, healthcare organizations can consider allocating additional resources to those regions. Recommendations should be practical, measurable, and supported by evidence from the analysis.

9. Why is effective presentation and data storytelling important in analytics projects?

Data storytelling helps communicate complex analytical results to both technical and non-technical audiences. Simply presenting tables and graphs may not clearly explain why a particular finding is important. A good data story combines relevant data, appropriate visualizations, and a logical explanation. It highlights important trends and connects them to real-world problems. Effective presentation helps stakeholders understand the results quickly and supports better data-driven decision-making.

10. Suggest future improvements or advanced techniques that could enhance the accuracy and effectiveness of the developed analytics solution.

The analytics solution can be improved using advanced techniques such as feature engineering, hyperparameter tuning, cross-validation, and ensemble learning. Algorithms such as Random Forest and Gradient Boosting can be tested against simpler models. For time-dependent COVID-19 data, time-series forecasting techniques can also be applied. More data can be collected to improve model generalization. Interactive dashboards can be developed using Power BI or Tableau. Finally, continuous model monitoring and periodic retraining can help maintain model performance when new data becomes available.